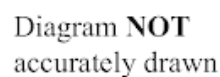


Nguyen Khac Kiem PhD

Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$



angle DAB = angle ABC = angle EFG = angle FGH = 90°

The angle of elevation of E from B is 50°

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

.....cm³

5 marks

2. 4MA1/1H_Winter_2025_Q19

The diagram shows triangle ABC

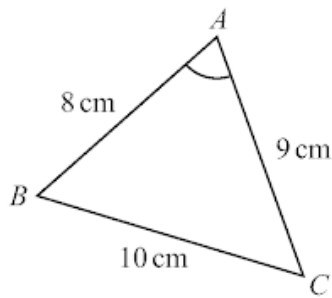


Diagram **NOT** accurately drawn

$AB = 8\text{ cm}$ $BC = 10\text{ cm}$ $CA = 9\text{ cm}$

Work out the size of angle BAC
Give your answer correct to one decimal place.

o

3 marks

3. 4MA1/1H_Winter_2025_Q15

AOB is a sector of a circle with centre O and radius 12 cm

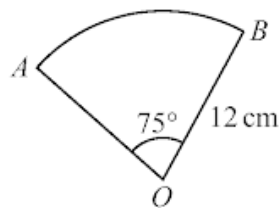


Diagram **NOT** accurately drawn

Work out the area of the sector.
Give your answer correct to 3 significant figures.

..... cm^2

2 marks

4. 4MA1/1H_Winter_2025_Q12

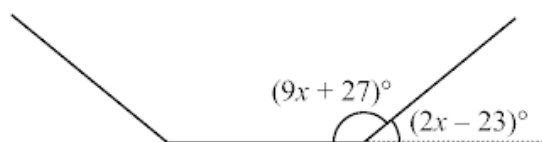


Diagram **NOT**
accurately drawn

The diagram shows part of a regular n -sided polygon with

an interior angle of $(9x + 27)^\circ$

an exterior angle of $(2x - 23)^\circ$

Work out the value of n

$n = \dots\dots\dots$

4 marks

5. 4MA1/1H_Winter_2025_Q11

ABC is a right-angled triangle.

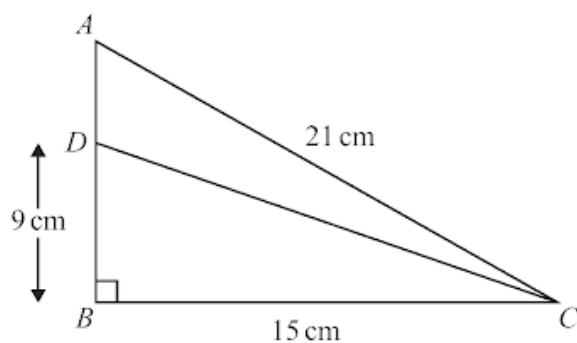


Diagram **NOT**
accurately drawn

$AC = 21 \text{ cm}$ $BC = 15 \text{ cm}$ angle $ABC = 90^\circ$

The point D lies on AB such that $DB = 9 \text{ cm}$

Work out the size of angle ACD

Give your answer correct to one decimal place.

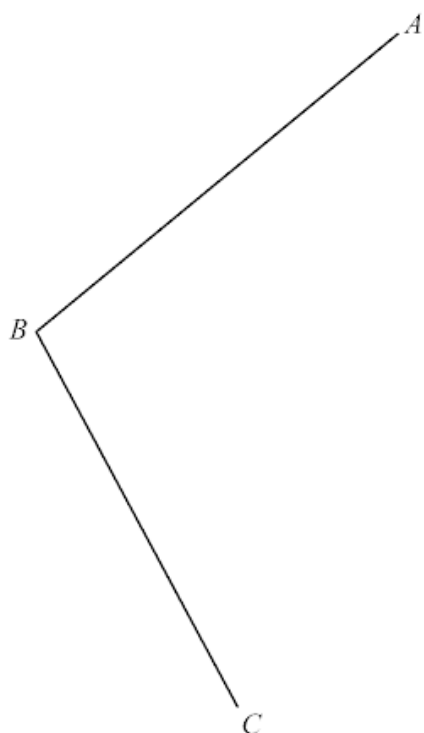
o

$\dots\dots\dots$

4 marks

6. 4MA1/1H_Winter_2025_Q3

Using ruler and compasses only, construct the bisector of angle ABC
Show all your construction lines.



2 marks

R, **S** and **T** are three similar vases.

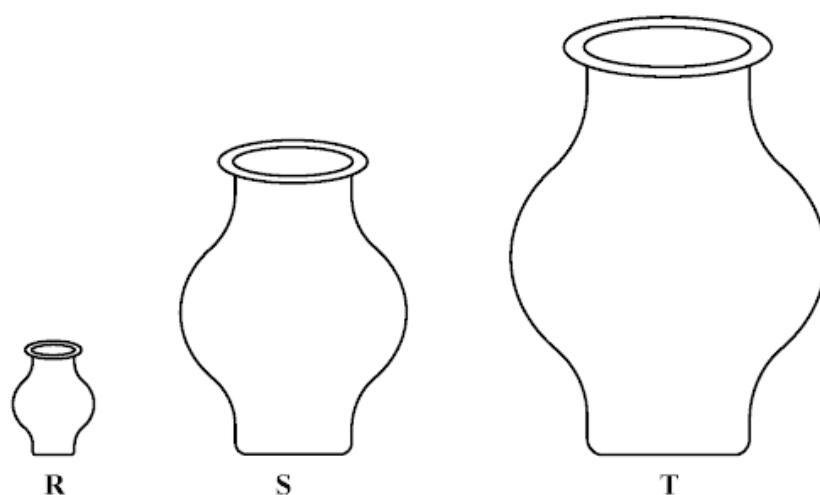


Diagram **NOT**
accurately drawn

The volume of vase **S** is 72.8% more than the volume of vase **R**

The height of vase **R** is h cm

The height of vase **T** is $6h$ cm

The surface area of vase **S** is A cm²

The surface area of vase **T** is kA cm²

Work out the value of k

$k = \dots\dots\dots$

4 marks

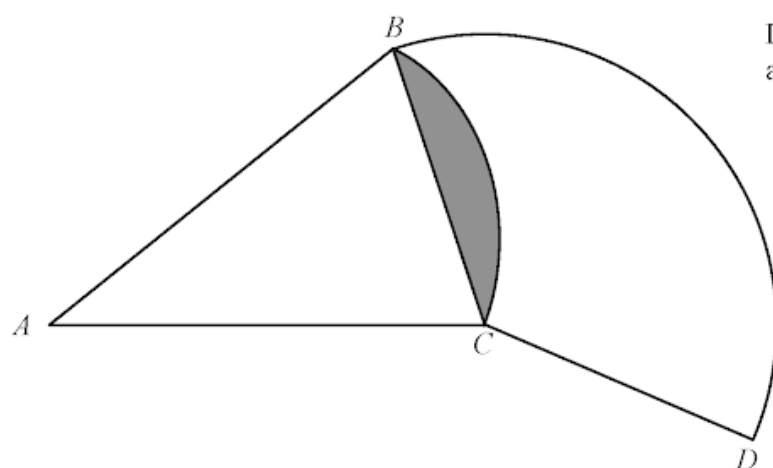


Diagram **NOT**
accurately drawn

BAC is a sector of a circle, centre A

BCD is a sector of a circle, centre C

Angle $BAC = 40^\circ$

Angle $BCD = 130^\circ$

Area of shaded segment = 28 cm^2

Find the length of the arc BD

Give your answer correct to 3 significant figures.

..... cm

6 marks

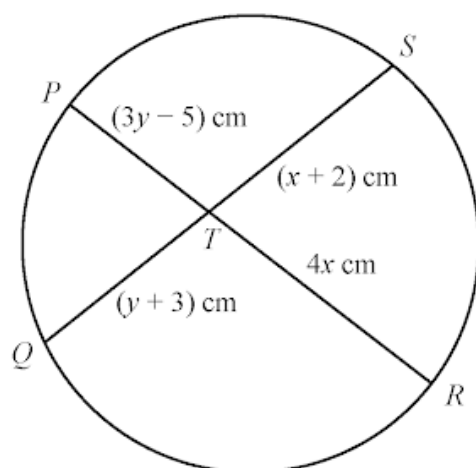


Diagram **NOT**
accurately drawn

PTR and QTS are chords of a circle.

$$PT = (3y - 5) \text{ cm} \quad QT = (y + 3) \text{ cm} \quad RT = 4x \text{ cm} \quad ST = (x + 2) \text{ cm}$$

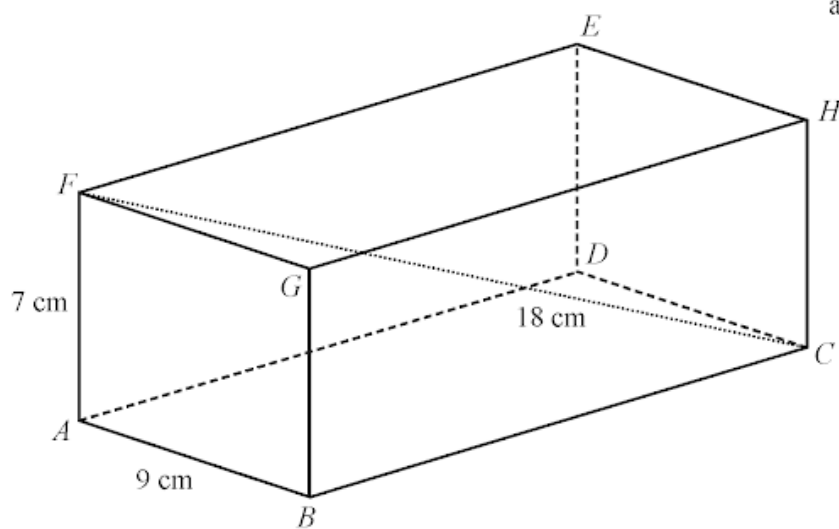
Find an expression for y in terms of x

$$y = \dots\dots\dots$$

5 marks

The diagram shows cuboid $ABCDEFGH$

Diagram **NOT**
accurately drawn



$$AB = 9 \text{ cm} \quad AF = 7 \text{ cm} \quad FC = 18 \text{ cm}$$

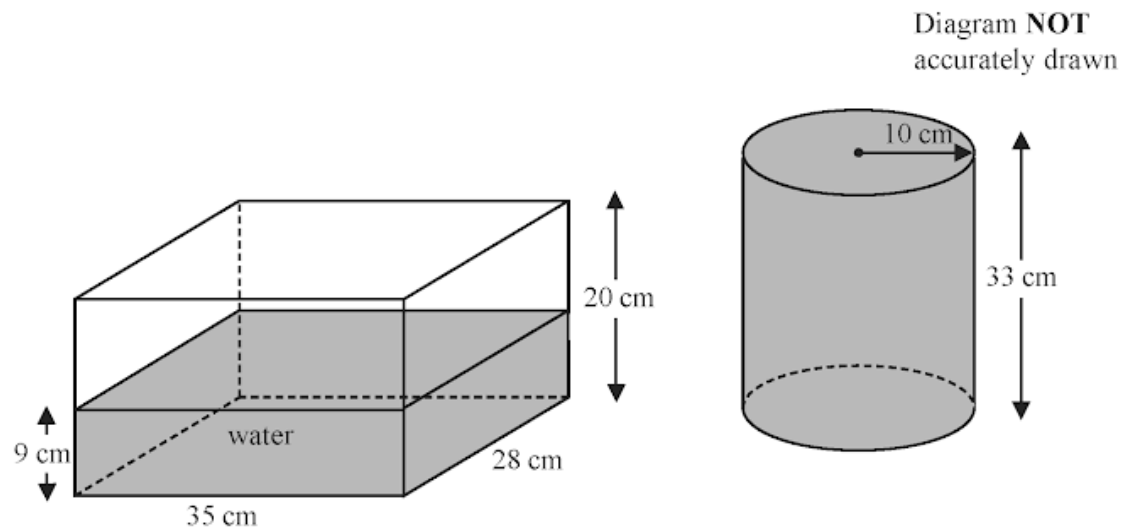
Calculate the length of BC

Give your answer correct to 3 significant figures.

..... cm

3 marks

The diagram shows two water containers.
One is a cuboid and one is a cylinder.



The cuboid measures 35 cm by 28 cm by 20 cm

The surface of the water in the cuboid is 9 cm above the base of the cuboid.

The cylinder has a radius of 10 cm and a height of 33 cm

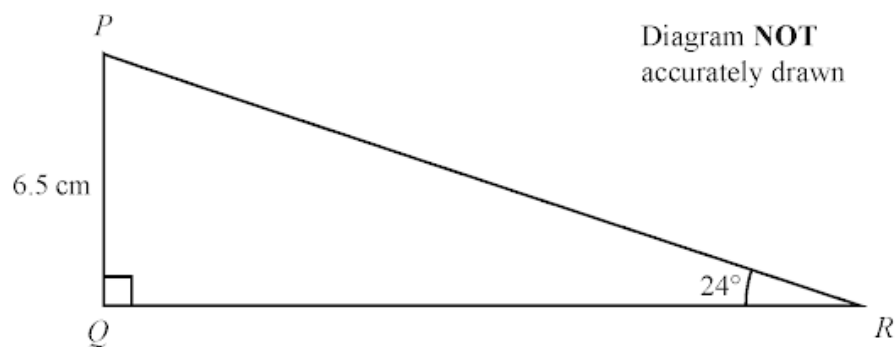
The cylinder is completely full of water.

Izzy is going to pour all the water from the cylinder into the cuboid.

Show that the cuboid will not be completely full of water.

3 marks

The diagram shows triangle PQR

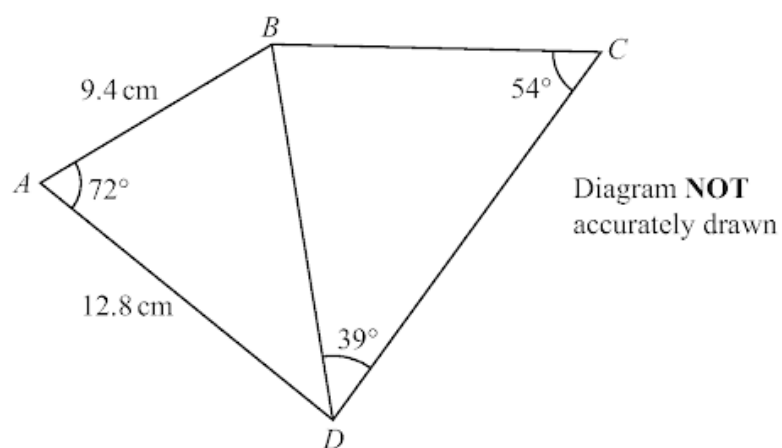


Work out the length of QR

Give your answer correct to 3 significant figures.

..... cm

3 marks



Work out the length of BC
 Give your answer correct to 3 significant figures.

..... cm

5 marks

$OABC$ is a sector of a circle, centre O

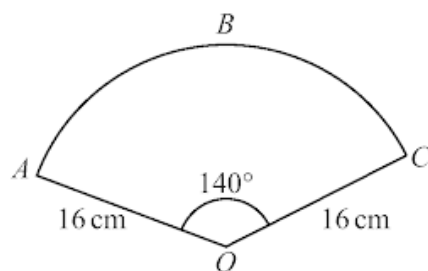


Diagram **NOT**
accurately drawn

angle $AOC = 140^\circ$

$OA = OC = 16\text{ cm}$

Calculate the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

2 marks

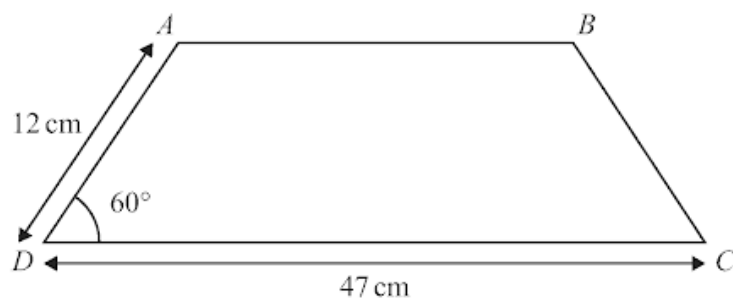


Diagram **NOT**
accurately drawn

$ABCD$ is a trapezium with one line of symmetry.

$$\text{angle } ADC = 60^\circ \quad AD = 12 \text{ cm} \quad DC = 47 \text{ cm}$$

Work out the area of the trapezium.

Give your answer correct to 3 significant figures.

Show your working clearly.

..... cm^2

5 marks

Here is a solid cylinder.

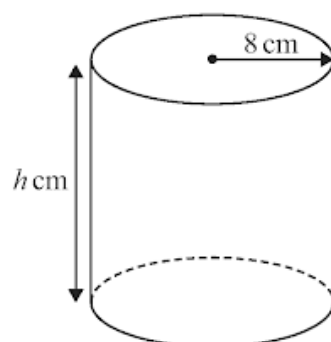


Diagram **NOT**
accurately drawn

The radius of the cylinder is 8 cm

The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h

Give your answer correct to one decimal place.

$h = \dots\dots\dots$

3 marks

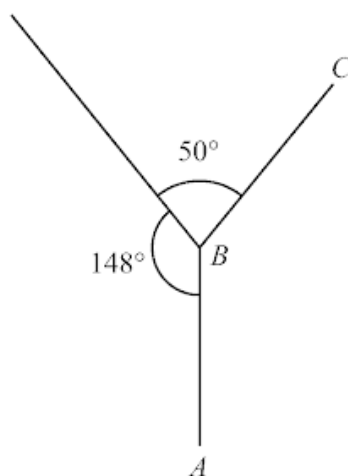


Diagram **NOT**
accurately drawn

AB and BC are two sides of a regular polygon with n sides.

Work out the value of n
Show your working clearly.

$n = \dots\dots\dots$

4 marks

The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

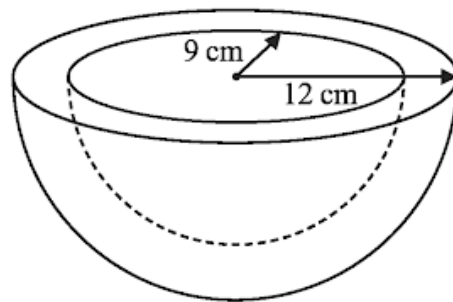


Diagram **NOT**
accurately drawn

The outer radius of the bowl is 12 cm
The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.
Give your answer correct to the nearest whole number.

3 marks

$OAPB$ is a sector of a circle, centre O

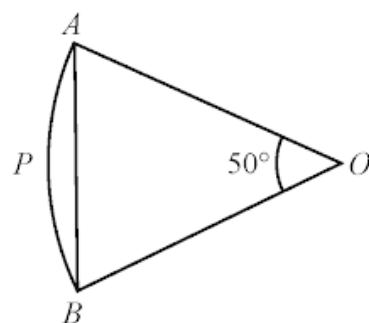


Diagram **NOT**
accurately drawn

Angle $AOB = 50^\circ$

Area of triangle $OAB = 120 \text{ cm}^2$

Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

4 marks

ABC is a right-angled triangle.

D is a point on BC

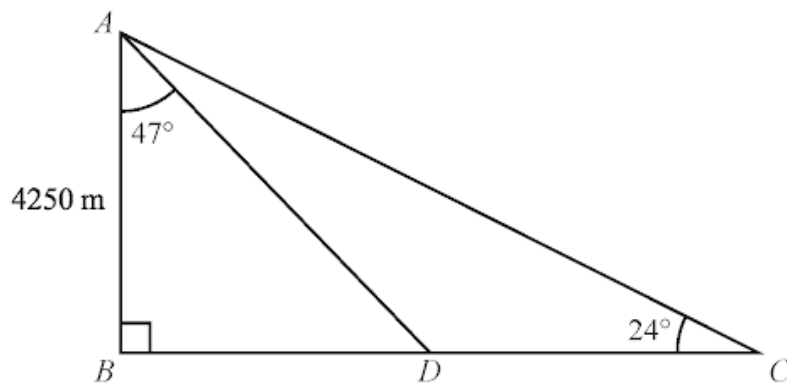


Diagram **NOT**
accurately drawn

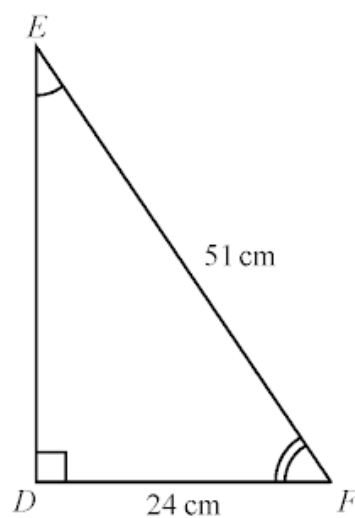
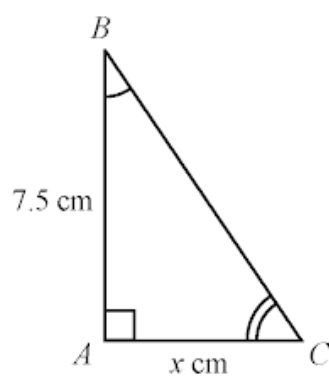
$AB = 4250 \text{ m}$ angle $BAD = 47^\circ$ angle $BCA = 24^\circ$

Work out the length of DC

Give your answer correct to the nearest integer.

4 marks

Here are two similar triangles.



Diagrams **NOT**
accurately drawn

Work out the value of x
Show your working clearly.

5 marks

The diagram shows a cylinder.

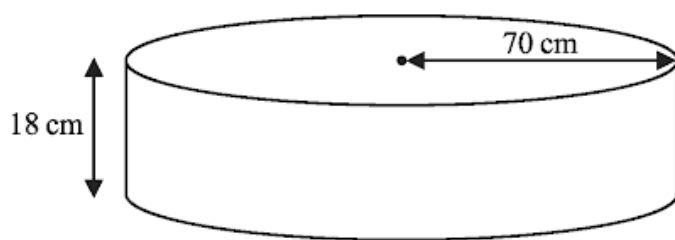


Diagram **NOT**
accurately drawn

The radius of the cylinder is 70 cm

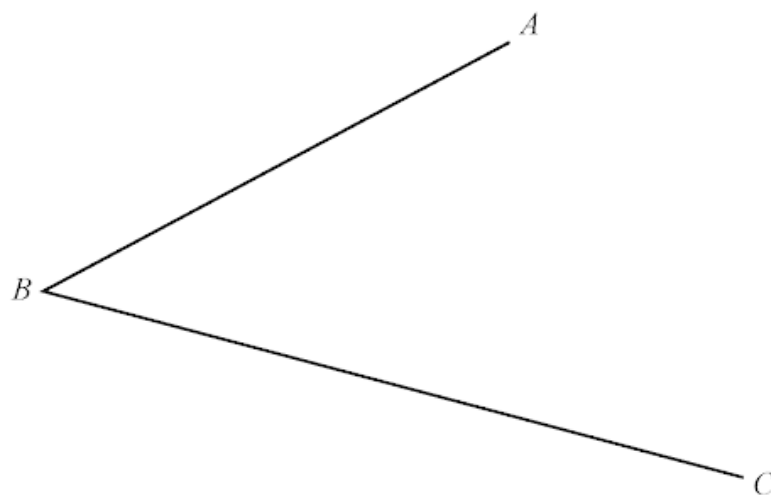
The height of the cylinder is 18 cm

Work out the volume of the cylinder.

Give your answer in litres correct to the nearest litre.

4 marks

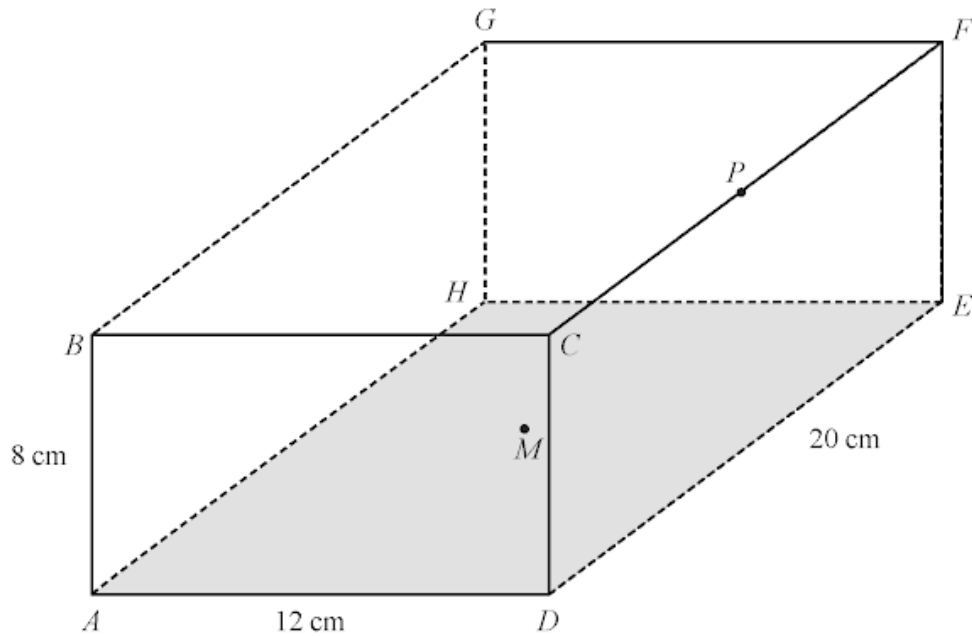
Use ruler and compasses only to construct the bisector of angle ABC
You must show all your construction lines.



2 marks

The diagram shows a cuboid $ABCDEFGH$ with horizontal base $ADEH$

Diagram **NOT** accurately drawn



$AB = 8\text{ cm}$ $AD = 12\text{ cm}$ $DE = 20\text{ cm}$

M is the midpoint of the base $ADEH$ and P is the midpoint of the edge CF

Work out the size of angle BMP
Give your answer correct to one decimal place.

.....°
(6)

The diagram shows two similar vases, **A** and **B**

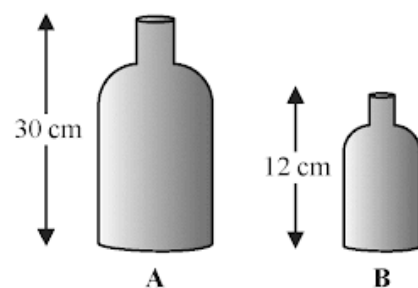


Diagram **NOT**
accurately drawn

The height of vase **A** is 30 cm

The height of vase **B** is 12 cm

Given that

$$\text{surface area of vase A} - \text{surface area of vase B} = 178.5 \text{ cm}^2$$

find the surface area of vase **A**

..... cm²

(4)

The diagram shows isosceles triangle EFG

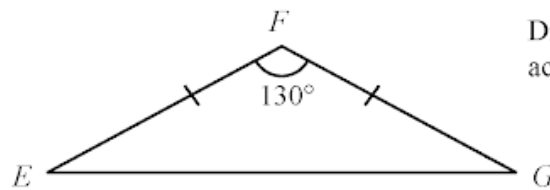


Diagram **NOT**
accurately drawn

$$EF = GF$$

$$\text{Angle } EFG = 130^\circ$$

The area of triangle EFG is 74 cm^2

Work out the length of EF

Give your answer correct to 3 significant figures.

..... cm

(3)

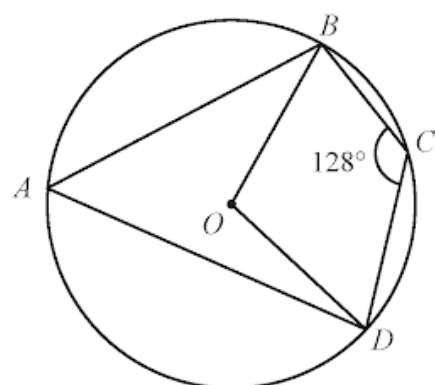


Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle, centre O

Angle $BCD = 128^\circ$

Work out the size of angle OBD

Give a reason for each stage of your working.

angle $OBD = \dots\dots\dots^\circ$

(5)

The diagram shows a trapezium, $ABCD$

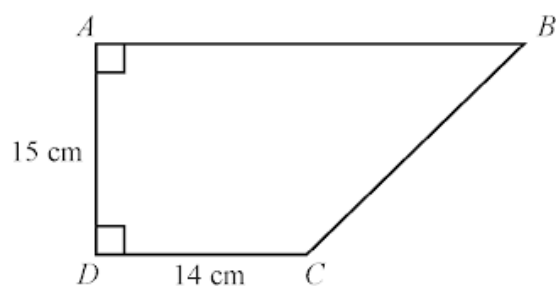


Diagram **NOT**
accurately drawn

DAB and ADC are right angles.

$$AD = 15\text{ cm} \quad DC = 14\text{ cm}$$

The area of the trapezium is 360 cm^2

Work out the perimeter of the trapezium.

..... cm

(6)

The diagram shows a solid wooden cylinder.

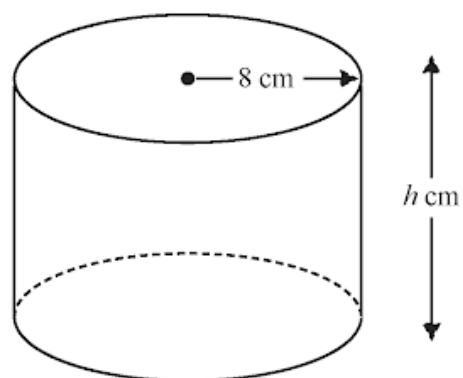


Diagram **NOT**
accurately drawn

The cylinder has radius 8 cm and height h cm.
The volume of the cylinder is 1208 cm^3

- (a) Work out the value of h
Give your answer correct to the nearest whole number.

(2)

The density of the wood is 1.25 g/cm^3

- (b) Work out the mass of the cylinder.
Give your answer in kilograms.

(2)

A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

o

(6)

A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

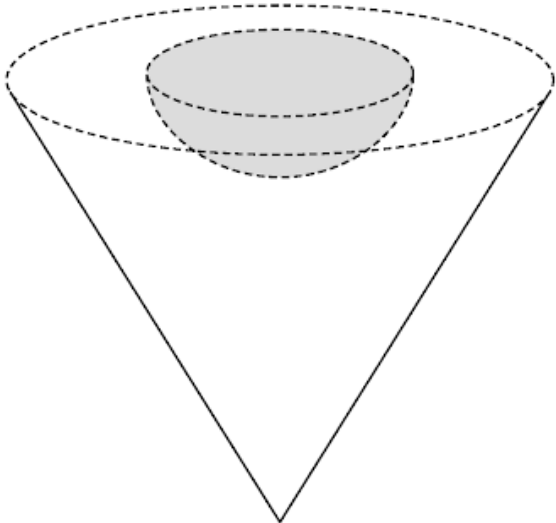


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x\text{ cm}$
The radius of the base of the cone is $5x\text{ cm}$
The vertical height of the cone is $6x\text{ cm}$

The volume of the solid shape is $6948\pi\text{ cm}^3$

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm^2

(5)

The diagram shows a sector $OABC$ of a circle centre O

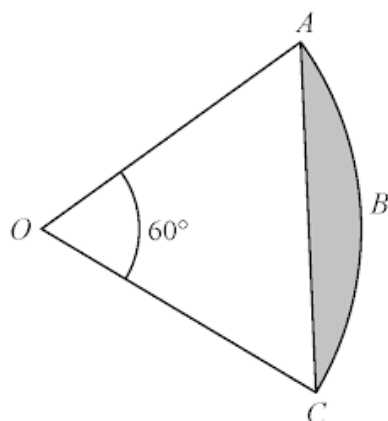


Diagram **NOT**
accurately drawn

Angle $AOC = 60^\circ$

The area of the shaded segment ABC is 38 cm^2

Work out the perimeter of the shaded segment ABC

Give your answer correct to one decimal place.

..... cm

(4)

Here is quadrilateral $ABCD$

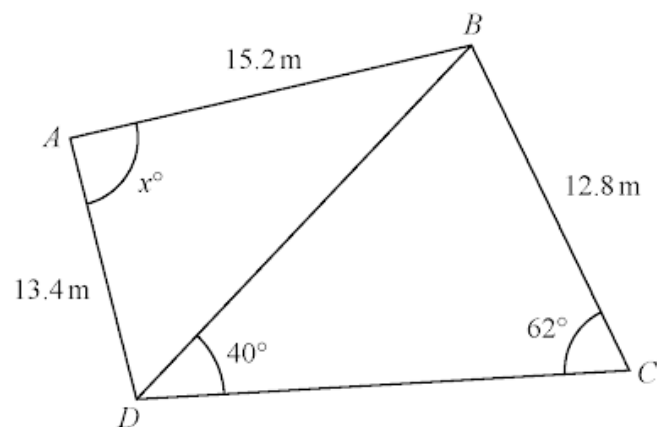


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

(5)

The diagram shows a roof support.

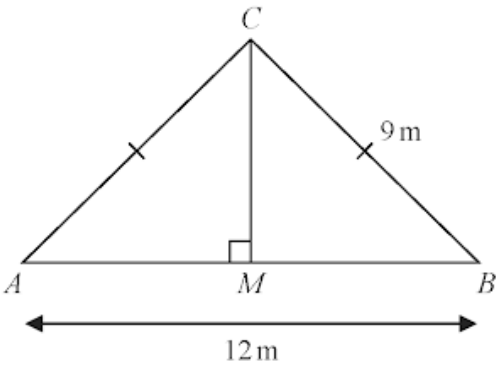


Diagram **NOT**
accurately drawn

The roof support is made from four lengths of wood, AB , AC , BC and MC

$AC = BC = 9\text{ m}$ $AB = 12\text{ m}$

angle $AMC = 90^\circ$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.
Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.
Show your working clearly.

..... euros

(4)

In the diagram, $ABCDE$ is a regular pentagon.

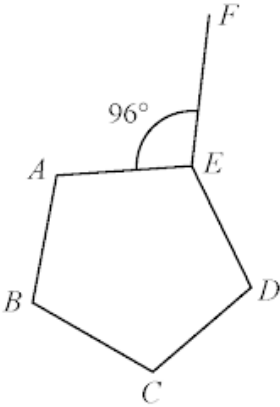


Diagram **NOT** accurately drawn

Angle $AEF = 96^\circ$

Work out the size of the obtuse angle FED
Show your working clearly.

o

.....

(4)

The diagram shows a triangular prism, $ABCDEF$, with a rectangular base $ABCD$

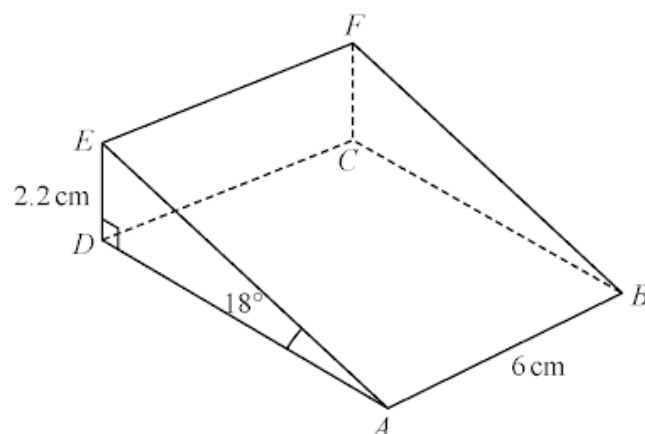


Diagram **NOT**
accurately drawn

$$AB = 6 \text{ cm}$$

$$DE = 2.2 \text{ cm}$$

$$\text{angle } DAE = 18^\circ$$

$$\text{angle } ADE = 90^\circ$$

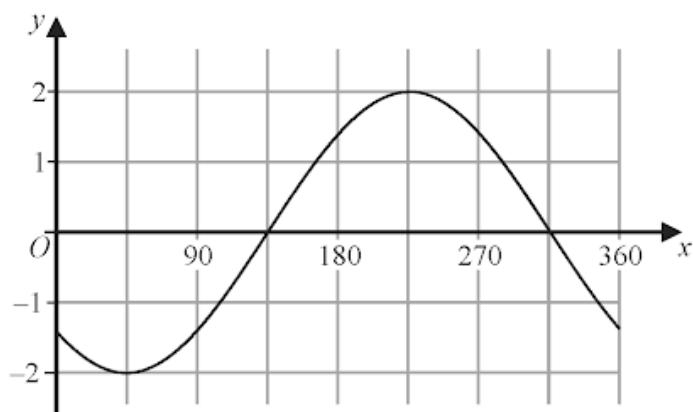
Work out the angle that BE makes with the plane $ABCD$

Give your answer correct to one decimal place.

○

4 marks

Here is a sketch of the curve $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$



Given that $0 < b < 180$

find the value of a and the value of b

$a =$

$b =$

2 marks

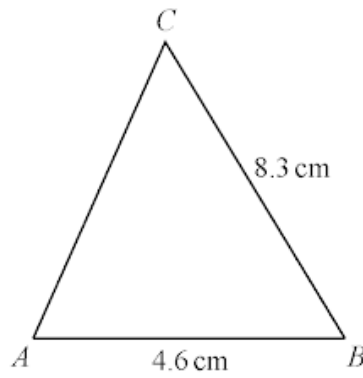


Diagram **NOT**
accurately drawn

$AB = 4.6 \text{ cm}$ $BC = 8.3 \text{ cm}$ angle ABC is acute

The area of triangle ABC is 12 cm^2

Work out the perimeter of triangle ABC

Give your answer correct to 3 significant figures.

..... cm

5 marks

R and T are points on a circle, centre O

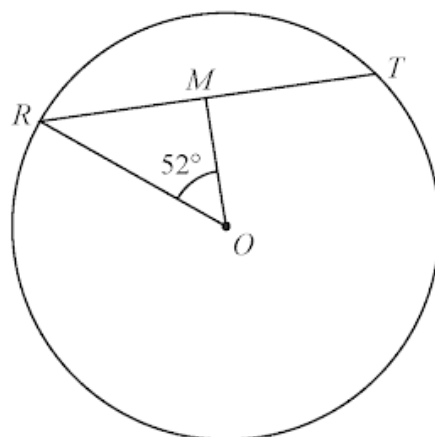


Diagram **NOT**
accurately drawn

$$RT = 12 \text{ cm}$$

M is the midpoint of RT

$$\text{Angle } ROM = 52^\circ$$

Work out the area of the circle.

Give your answer correct to 3 significant figures.

..... cm^2

4 marks

The diagram shows an isosceles triangle ABC

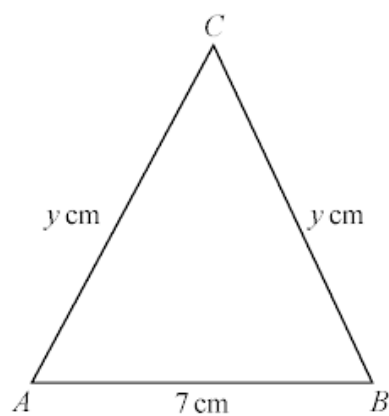


Diagram **NOT**
accurately drawn

$$AB = 7\text{ cm} \quad AC = BC = y\text{ cm}$$

The area of the triangle is 42 cm^2

Work out the value of y

$$y = \dots\dots\dots$$

4 marks

The diagram shows a solid triangular prism.

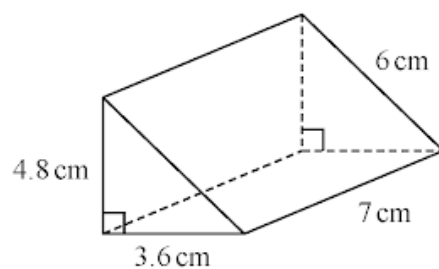


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the triangular prism.
Give your answer correct to 3 significant figures.

.....cm²

3 marks

The diagram shows the cross section of a circular water pipe.

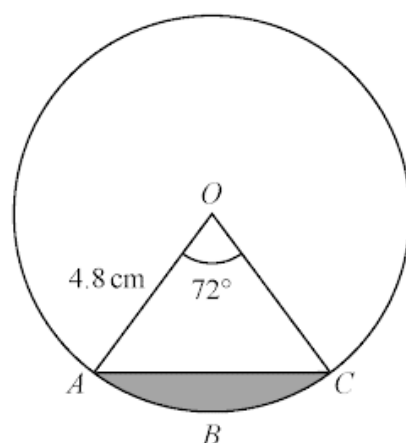


Diagram **NOT**
accurately drawn

$OABC$ is a sector of the circle, centre O

The shaded region in the diagram represents the water flowing in the pipe.

The water flows at 14 cm/s in the pipe.

Work out the volume of water that has flowed through the pipe in 3 minutes.
Give your answer in cm^3 correct to 3 significant figures.

Here is a shape formed from two triangles ABC and CDE
 ACD and BCE are straight lines.

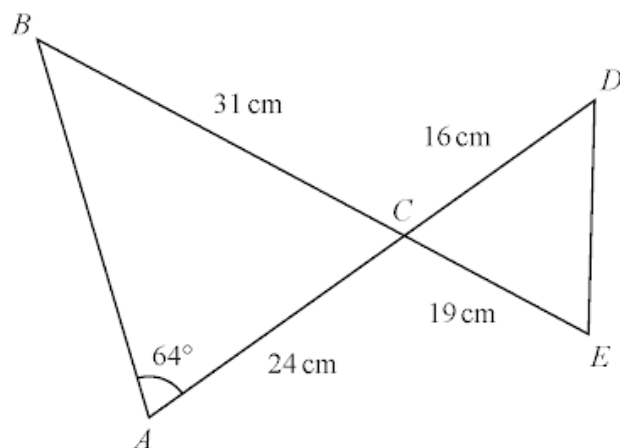


Diagram **NOT**
accurately drawn

$$AC = 24 \text{ cm} \quad BC = 31 \text{ cm} \quad CE = 19 \text{ cm} \quad CD = 16 \text{ cm}$$

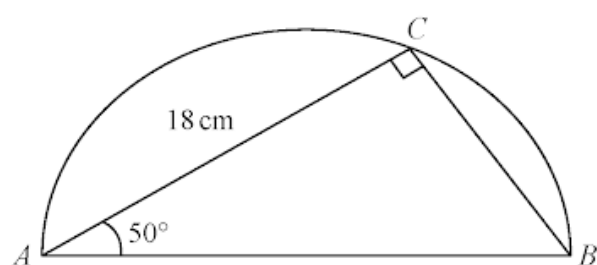
$$\text{Angle } BAC = 64^\circ$$

Work out the length of DE

Give your answer correct to 3 significant figures.

The diagram shows a triangle ABC inside a semicircle.

Diagram **NOT**
accurately drawn



A , B and C are points on the semicircle.

AB is the diameter of the semicircle.

Angle $ACB = 90^\circ$

Angle $BAC = 50^\circ$

$AC = 18\text{ cm}$

Work out the perimeter of the semicircle.

Give your answer correct to 2 significant figures.

The diagram shows a pentagon.

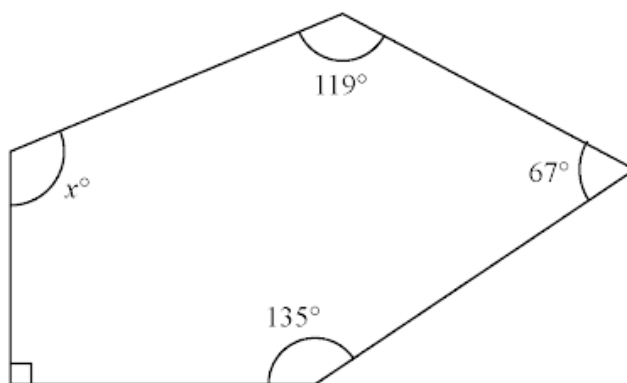


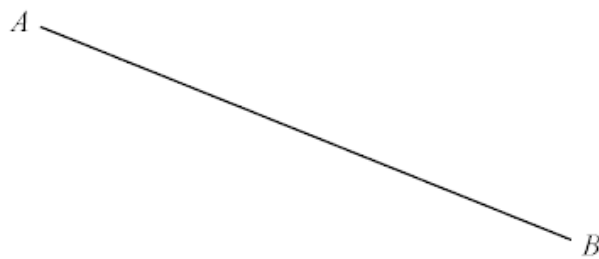
Diagram **NOT**
accurately drawn

Work out the value of x

$x =$

3 marks

Use ruler and compasses only to construct the perpendicular bisector of line AB
You must show all your construction lines.



2 marks

Here is a triangle ABC

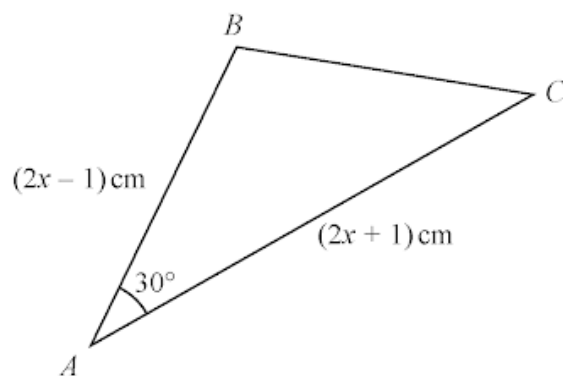


Diagram **NOT**
accurately drawn

The area of the triangle is $(x^2 + x - 3.75) \text{ cm}^2$

Find the size of the largest angle in triangle ABC

Give your answer correct to the nearest degree.

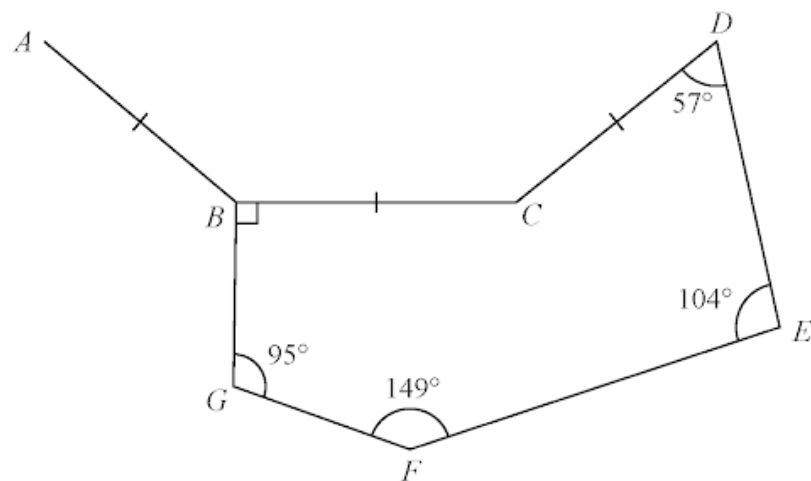


Diagram **NOT**
accurately drawn

$BCDEFG$ is a hexagon.

AB , BC and CD are three sides of a regular n -sided polygon.

Calculate the value of n

Show your working clearly.

$$n = \dots\dots\dots$$

4 marks

The diagram shows a block of iron in the shape of a cuboid.

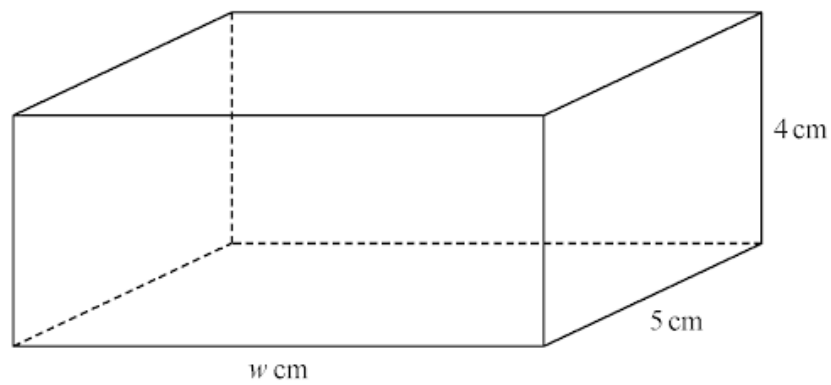


Diagram **NOT**
accurately drawn

The block has length w cm, width 5 cm and height 4 cm

The density of iron is 7.8 g/cm^3

The mass of the block is 1950 g

Work out the value of w

$w =$

3 marks

The diagram shows a shape made up of three semicircles, enclosing a right-angled triangle.

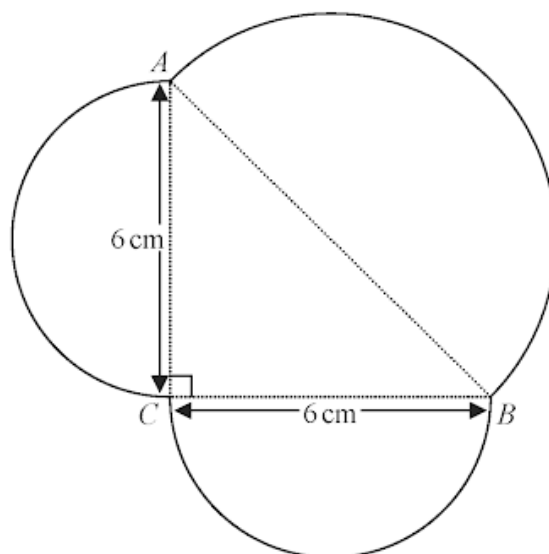


Diagram **NOT**
accurately drawn

AB , BC and CA are each the diameter of a semicircle.

$BC = CA = 6$ cm.

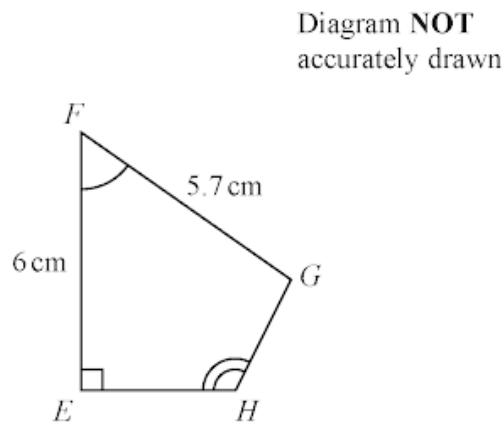
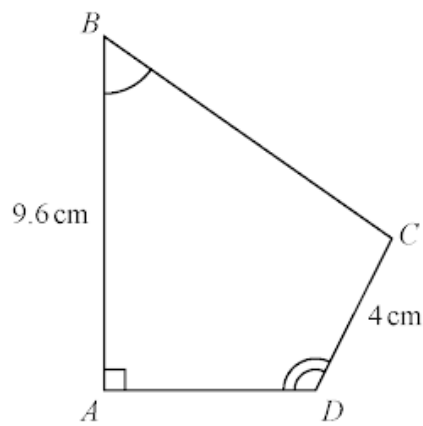
Work out the perimeter of the shape.

Give your answer correct to one decimal place.

..... cm

5 marks

$ABCD$ and $EFGH$ are similar quadrilaterals.



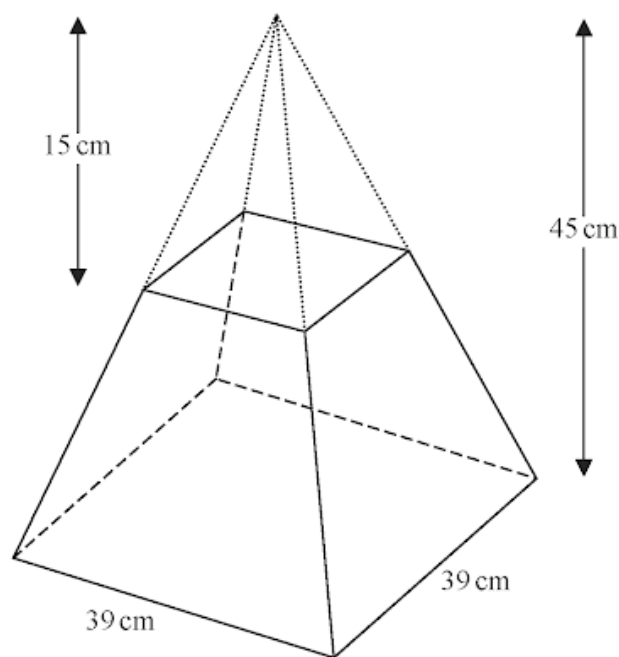
(a) Work out the length of GH

..... cm
(2)

(b) Work out the length of BC

..... cm
(2)

A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.



The height of the small pyramid is 15 cm.

The height of the large pyramid is 45 cm.

The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.

Give your answer correct to the nearest whole number.

The diagram shows two circles with centre O and a regular pentagon $ABCDE$

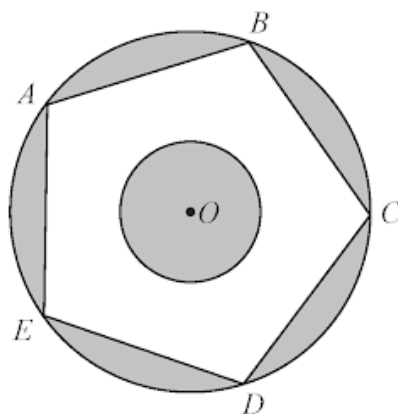


Diagram **NOT**
accurately drawn

A, B, C, D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

The diagram shows a shaded shape $AEBCD$ made by removing triangle AEB from rectangle $ABCD$

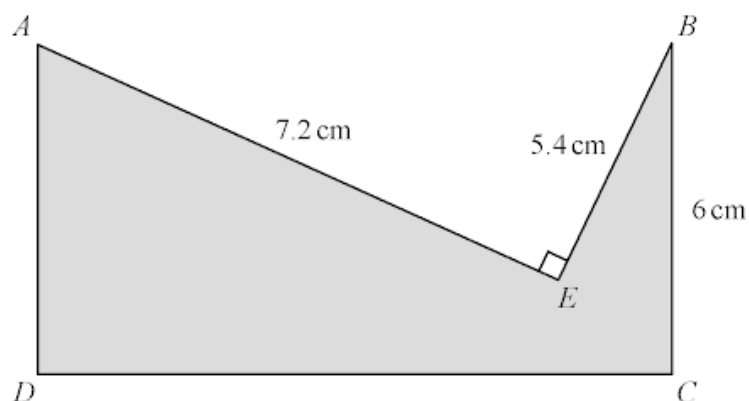


Diagram **NOT**
accurately drawn

$$AE = 7.2 \text{ cm} \quad BE = 5.4 \text{ cm} \quad BC = 6 \text{ cm} \quad \text{angle } AEB = 90^\circ$$

Work out the perimeter of the shaded shape.

..... cm

4 marks

The diagram shows triangle ABC

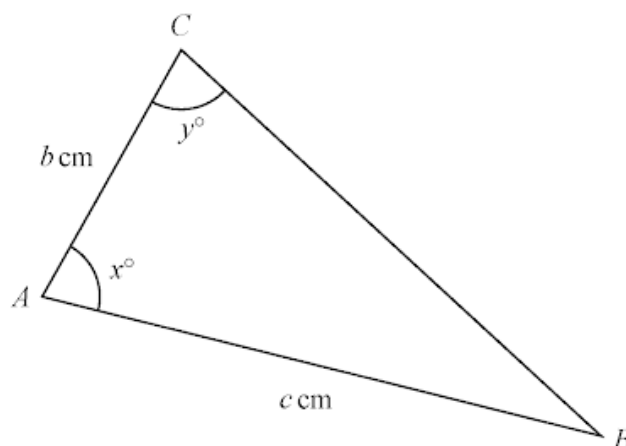


Diagram **NOT**
accurately drawn

$c = 11.5$ correct to one decimal place
 $x = 80$ correct to the nearest whole number
 $y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of b
Show your working clearly.
Give your answer correct to 3 significant figures.

.....
4 marks

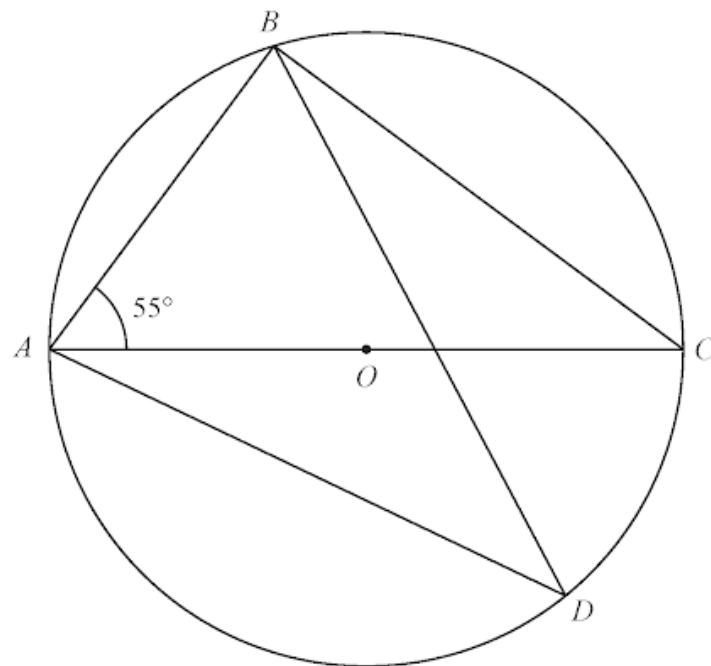


Diagram **NOT**
accurately drawn

A , B , C and D are points on a circle, centre O
 AOC is a diameter of the circle.

Angle $BAC = 55^\circ$

Work out the size of angle ADB

Give a reason for each stage of your working.

◇

4 marks

The diagram shows sector OPQ of a circle, centre O

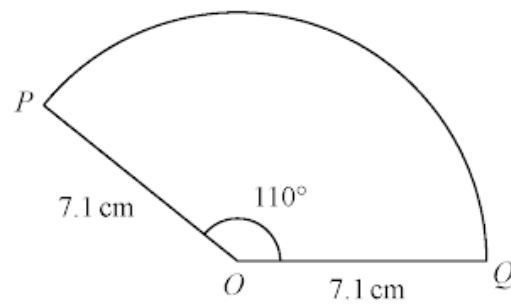


Diagram **NOT**
accurately drawn

$$OP = OQ = 7.1 \text{ cm}$$

$$\text{Angle } POQ = 110^\circ$$

Calculate the area of sector OPQ

Give your answer correct to one decimal place.

..... cm^2

2 marks

The shaded shape is made using three identical right-angled triangles and a square.

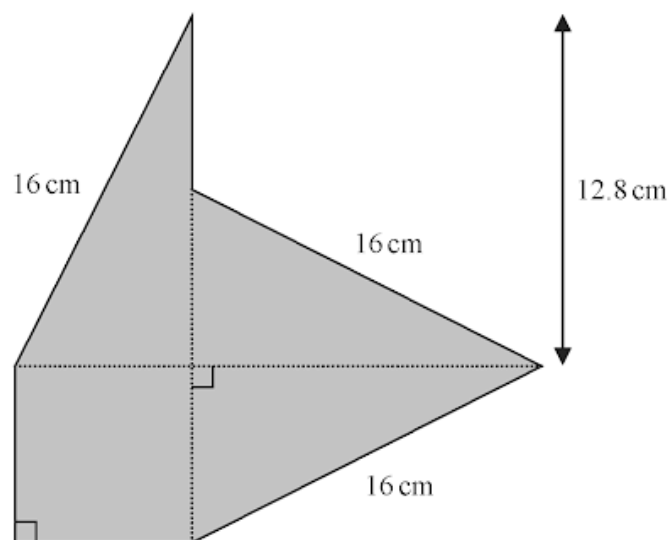


Diagram **NOT**
accurately drawn

Work out the perimeter of the shaded shape.

..... cm

4 marks

The diagram shows the prism $ABCDEF GHJK$ with horizontal base $AEFG$

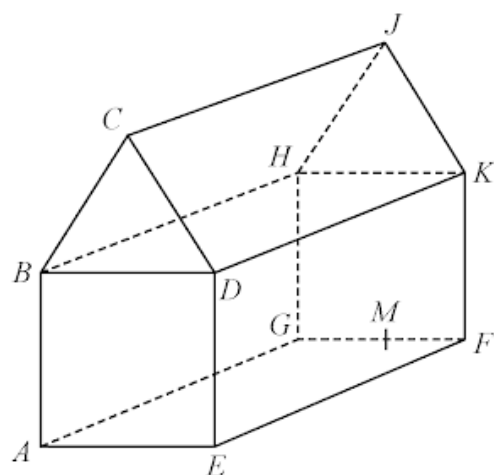


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where

$ABDE$ is a square

BCD is an equilateral triangle

$$EF = 2 \times AE$$

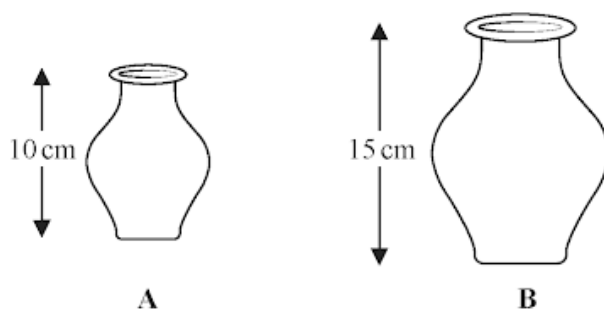
M is the midpoint of GF so that JM is vertical.

Angle $MAJ = y^\circ$

Given that $\tan y^\circ = T$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

A and **B** are two similar vases.



Vase **A** has height 10 cm.

Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

Calculate the volume of vase **A**

..... cm^3

4 marks

The diagram shows a circle with centre O

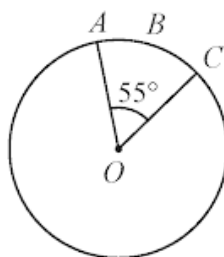


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

4 marks

The diagram shows isosceles triangle ABC

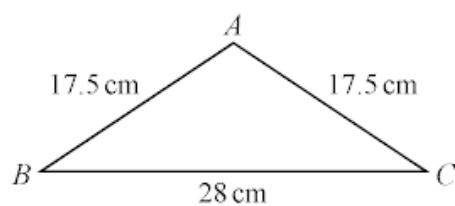


Diagram **NOT**
accurately drawn

$$AB = AC = 17.5 \text{ cm}$$

$$BC = 28 \text{ cm}$$

Calculate the area of triangle ABC

..... cm^2

4 marks

The diagram shows triangle ABC

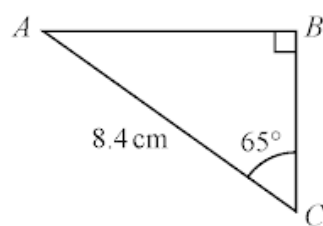


Diagram **NOT**
accurately drawn

Work out the length of the side AB
Give your answer correct to 3 significant figures.

..... cm

3 marks

The diagram shows two similar metal statues.



A



B

Diagram **NOT**
accurately drawn

The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is $k\%$ less than the surface area of statue **A**

Work out the value of k

Give your answer correct to 3 significant figures.

$k = \dots\dots\dots$

4 marks

The diagram shows a cube $ABCDEFGH$ with sides of length 6 cm.

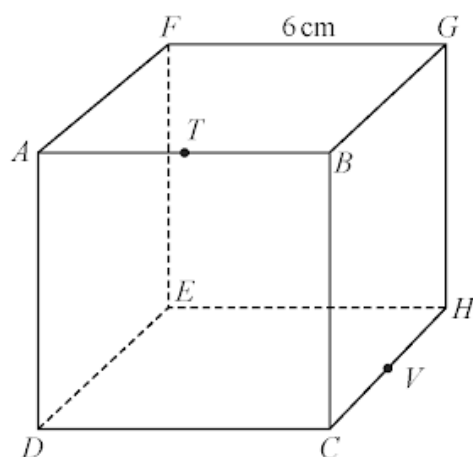


Diagram **NOT**
accurately drawn

T is the midpoint of AB and V is the midpoint of CH

Work out the distance from T to V in a straight line through the cube.
Give your answer in the form $\sqrt{a}$ cm where a is an integer.

..... cm

4 marks

Here is a parallelogram $PQRS$, in which angle SPQ is acute.

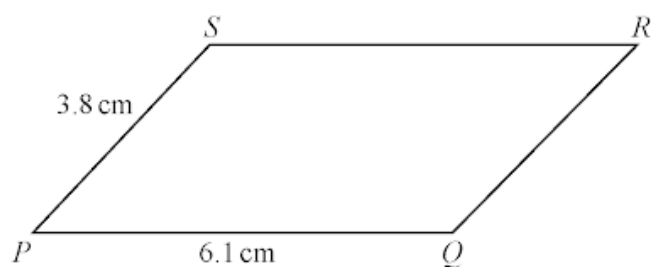


Diagram **NOT**
accurately drawn

$$PQ = 6.1\text{ cm} \quad PS = 3.8\text{ cm}$$

The area of the parallelogram is 18 cm^2

Work out the length of QS

Give your answer correct to 3 significant figures.

..... cm

5 marks

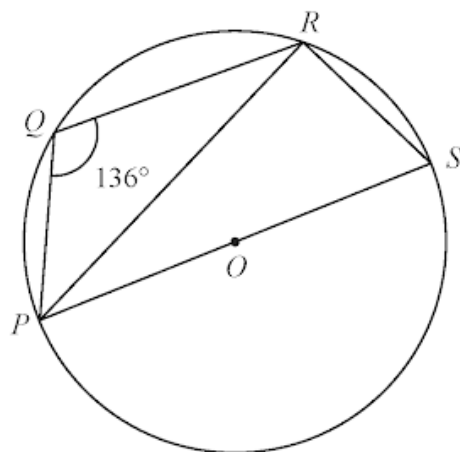


Diagram **NOT**
accurately drawn

P , Q , R and S are points on a circle with centre O

PS is a diameter of the circle.

Angle $PQR = 136^\circ$

Work out the size of angle RPS

○

3 marks

The diagram shows a quadrilateral $ABCD$

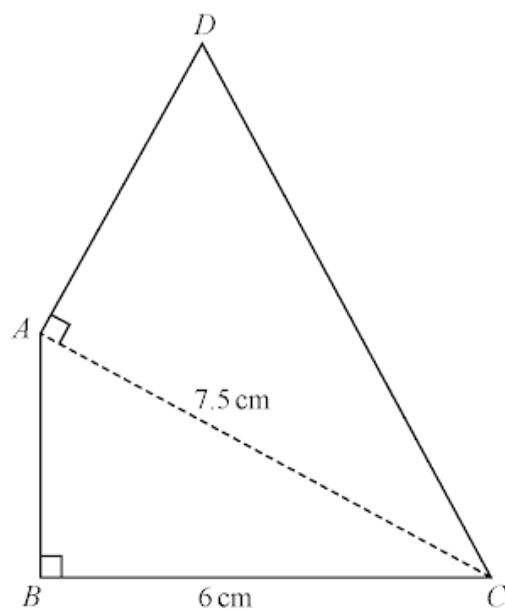


Diagram **NOT**
accurately drawn

In the diagram, ABC and DAC are right-angled triangles.

$$BC = 6 \text{ cm} \quad AC = 7.5 \text{ cm}$$

The area of quadrilateral $ABCD$ is 31.5 cm^2

Work out the length of AD

The diagram shows a solid cylinder made from iron.

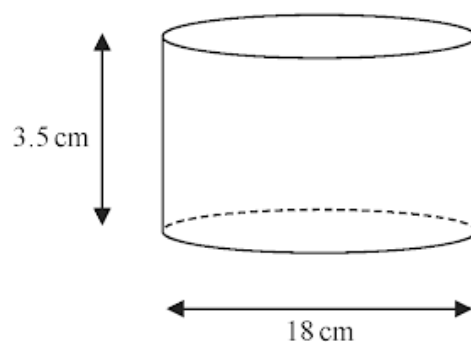


Diagram **NOT**
accurately drawn

The cylinder has diameter 18 cm and height 3.5 cm

The mass of the cylinder is 7.04 kg

Work out the density of the iron.

Give your answer in g/cm^3 correct to 2 significant figures.

..... g/cm^3

3 marks

A solid is made from a cone and a hemisphere.

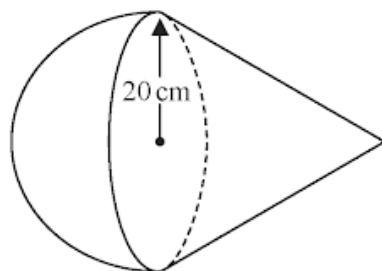


Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

$$k = \dots\dots\dots$$

5 marks

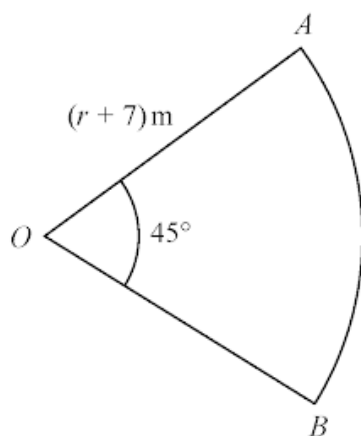


Diagram **NOT**
accurately drawn

OAB is a sector **S** of a circle with centre O and radius $(r + 7)$ metres.
Angle $AOB = 45^\circ$

A circle **C** has radius $(r - 2)$ metres.

The area of sector **S** is twice the area of circle **C**

Find the value of r

Show your working clearly.

Here is triangle ABC

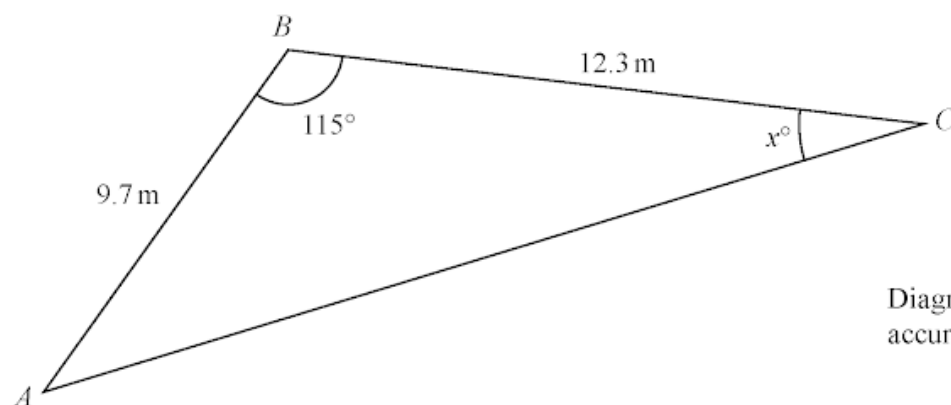


Diagram **NOT**
accurately drawn

Work out the value of x

Give your answer correct to 3 significant figures.

$x =$

5 marks

Two circles, C_1 and C_2 , are drawn on a centimetre grid, with a scale of 1 cm for 1 unit on each axis.

The centre of circle C_1 is at the point with coordinates $(-1, 3)$ and the radius of C_1 is 13 cm.

The centre of circle C_2 is at the point with coordinates $(7, 18)$ and the radius of C_2 is 6 cm.

(a) Work out the distance between the centre of C_1 and the centre of C_2

..... cm
(3)

(b) Explain why circle C_1 intersects circle C_2

.....

.....

(1)

.....

.....

.....

.....

.....

A cylinder is placed on the ground.

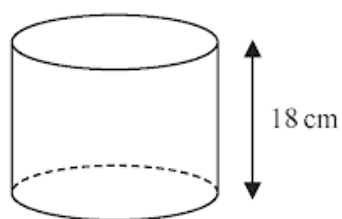


Diagram **NOT**
accurately drawn

The height of the cylinder is 18 cm.

The force exerted by the cylinder on the ground is 72 newtons.

The pressure on the ground due to the cylinder is 1.4 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cylinder.

Give your answer correct to 3 significant figures.

..... cm³

4 marks

The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

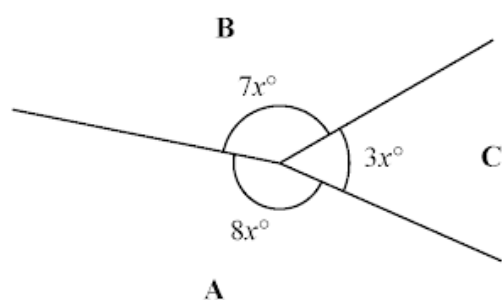


Diagram **NOT**
accurately drawn

Polygon **B** has n sides.

Work out the value of n .

$$n = \dots\dots\dots$$

4 marks

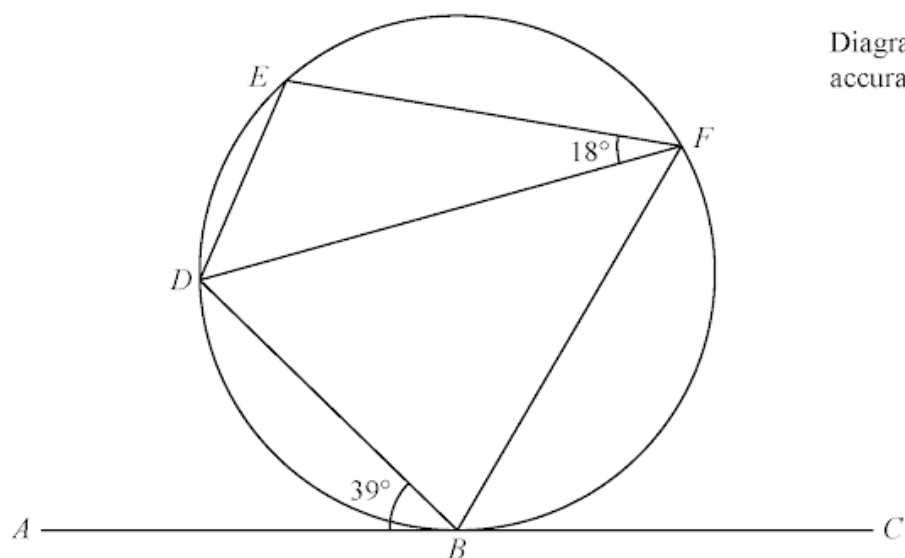


Diagram **NOT**
accurately drawn

B , D , E and F are points on a circle.

ABC is the tangent at B to the circle.

Angle $ABD = 39^\circ$

Angle $EFD = 18^\circ$

Work out the size of angle BDE .

Give reasons for your working.

o

(Total for Question is 4 marks)

The diagram shows a regular pentagon, $ABCDE$, a regular hexagon, $CFGHID$, and a quadrilateral, $EDIJ$.

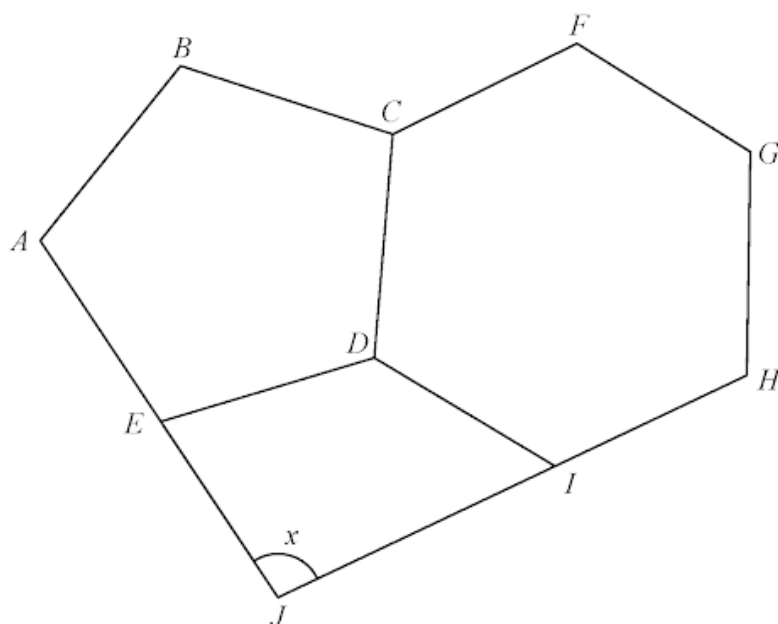


Diagram **NOT**
accurately drawn

AEJ and HIJ are straight lines.

Work out the size of the angle marked x .
Show your working clearly.

o

(Total for Question is 5 marks)

The diagram shows a cuboid and a cylinder.

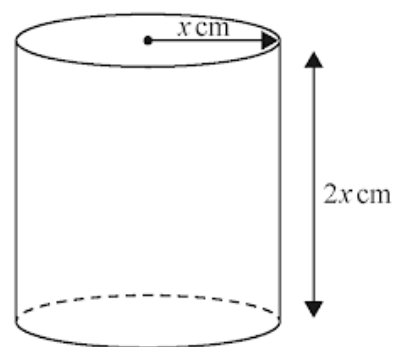
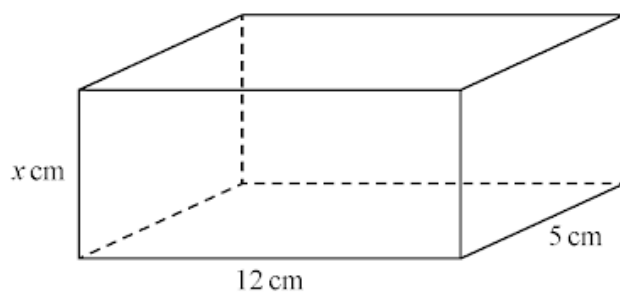


Diagram **NOT**
accurately drawn

The dimensions of the cuboid are $x\text{ cm}$ by 12 cm by 5 cm .
The volume of the cuboid is 270 cm^3

The radius of the cylinder is $x\text{ cm}$.
The height of the cylinder is $2x\text{ cm}$.

- (a) Work out the volume of the cylinder.
Give your answer correct to the nearest whole number.

..... cm^3
(3)

- (b) Change 1 m^3 to cm^3

..... cm^3
(1)

A and **B** are two similar solids.

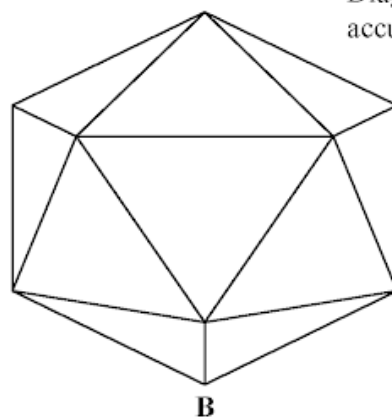
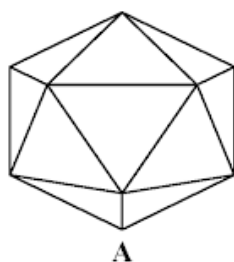


Diagram **NOT**
accurately drawn

A has a volume of 1836 cm^3

B has a volume of 4352 cm^3

B has a total surface area of 1120 cm^2

Work out the total surface area of **A**.

..... cm^2

(Total for Question is 3 marks)

A solid, **S**, is made from a hemisphere and a cylinder.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.

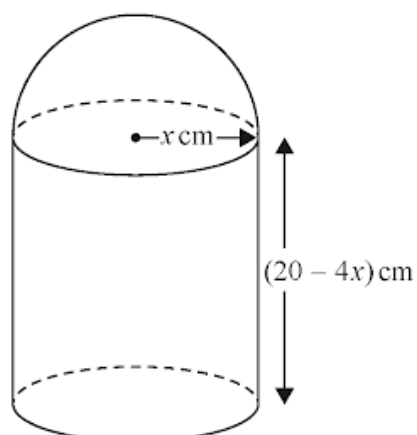


Diagram **NOT**
accurately drawn

The radius of the cylinder and the radius of the hemisphere are both x cm.

The height of the cylinder is $(20 - 4x)$ cm.

The volume of **S** is V cm³ where $V = \frac{1}{3}\pi y$

Find the maximum value of y .

Show clear algebraic working.

The diagram shows the positions of three ships, A , B and C .

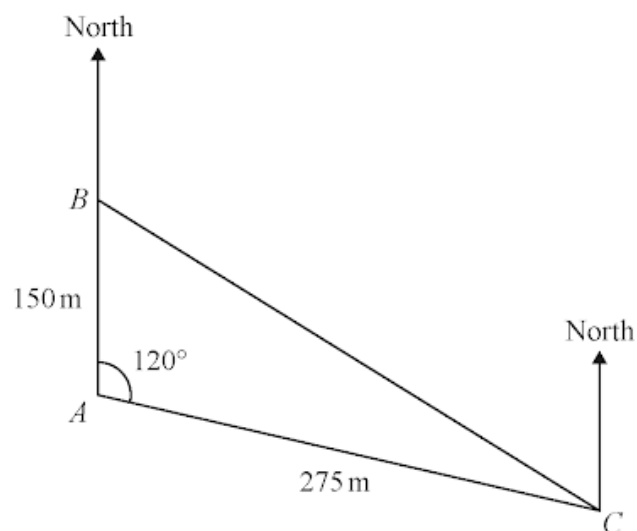


Diagram **NOT**
accurately drawn

Ship B is due north of ship A .

The bearing of ship C from ship A is 120°

Calculate the bearing of ship C from ship B .

Give your answer correct to the nearest degree.

Here is isosceles triangle ABC .

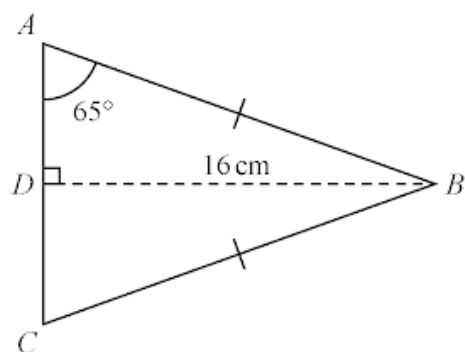


Diagram **NOT**
accurately drawn

D is the midpoint of AC and $DB = 16$ cm.

Angle $DAB = 65^\circ$

Work out the perimeter of triangle ABC .

Give your answer correct to one decimal place.

..... cm

(Total for Question is 4 marks)

The region, shown shaded in the diagram, is a path.

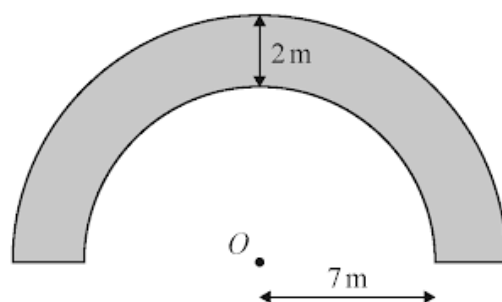


Diagram **NOT**
accurately drawn

The boundary of the path is formed by two semicircles, with the same centre O , and two straight lines.

The inner semicircle has a radius of 7 metres.

The path has a width of 2 metres.

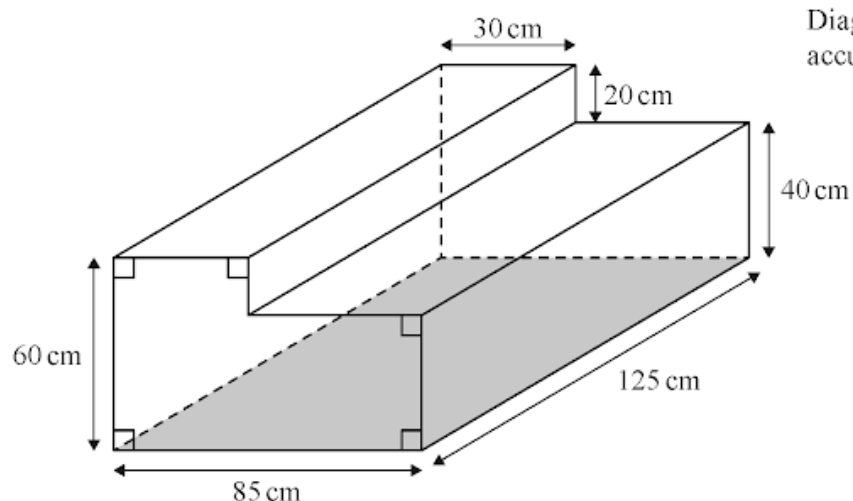
Work out the perimeter of the path.

Give your answer correct to one decimal place.

..... m

(Total for Question is 3 marks)

The diagram shows a container for water in the shape of a prism.



The rectangular base of the prism, shown shaded in the diagram, is horizontal.
The container is completely full of water.

Tuah is going to use a pump to empty the water from the container so that the volume of water in the container decreases at a constant rate.

The pump starts to empty water from the container at 1030 and at 1200 the water level in the container has dropped by 20 cm.

Find the time at which all the water has been pumped out of the container.

(Total for Question is 4 marks)

The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

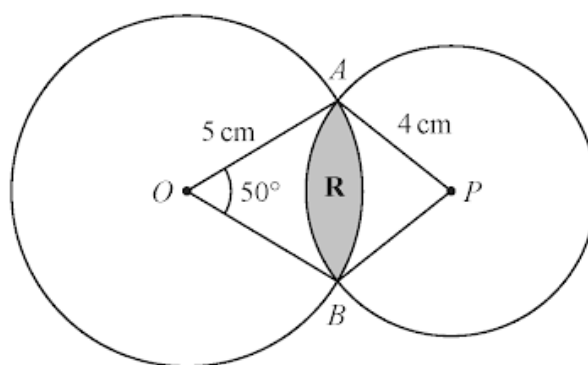


Diagram **NOT**
accurately drawn

One of the circles has centre O and radius 5 cm.

The other circle has centre P and radius 4 cm.

Angle $AOB = 50^\circ$

Calculate the area of region **R**.

Give your answer correct to 3 significant figures.

The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

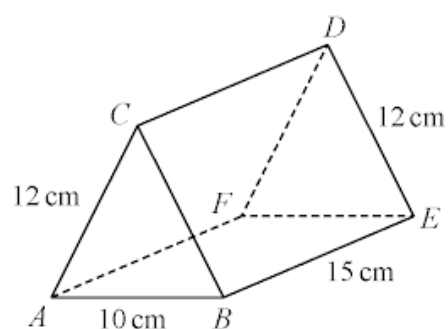


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm}$$

$$AB = 10 \text{ cm}$$

$$BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.

Give your answer correct to 3 significant figures.

4 marks

P , Q and R are points on a circle, centre O .
 TRV is the tangent to the circle at R .

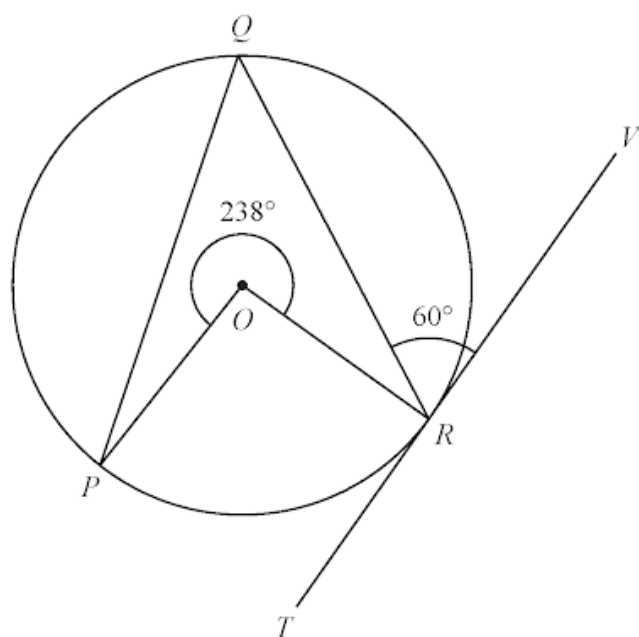


Diagram **NOT**
 accurately drawn

Reflex angle $POR = 238^\circ$

Angle $QRV = 60^\circ$

Calculate the size of angle OPQ .

Give a reason for each stage of your working.

4 marks

The diagram shows a sector of a circle with radius 7 cm.

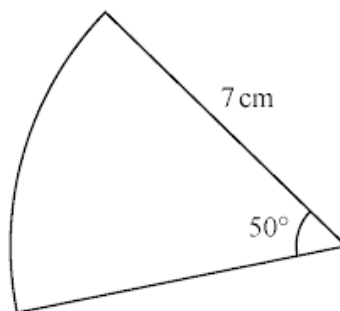
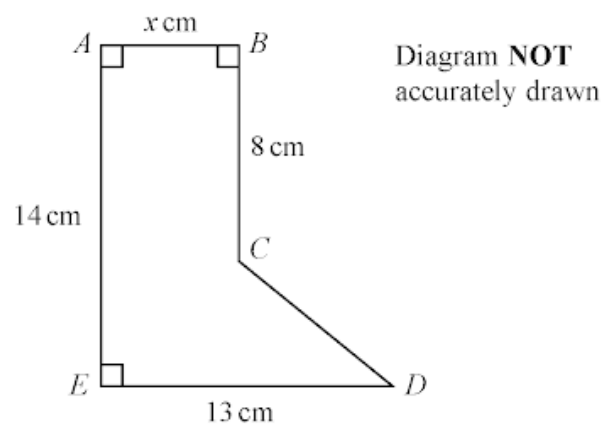


Diagram **NOT**
accurately drawn

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

..... cm

2 marks



The diagram shows the shape $ABCDE$.

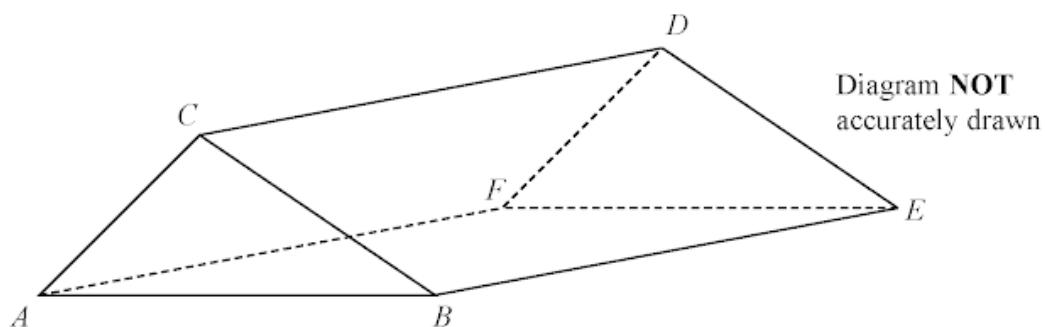
The area of the shape is 91.8 cm^2

Work out the value of x .

$$x = \dots\dots\dots$$

4 marks

The diagram shows the prism $ABCDEF$ with cross section triangle ABC .



Angle $BEC = 40^\circ$ and angle ACB is obtuse.
 $AC = 6$ cm and $CE = 13$ cm

The area of triangle ABC is 22 cm^2

Calculate the length of AB .

Give your answer correct to one decimal place.

..... cm

(Total for Question is 6 marks)

The diagram shows a frustum of a cone and a sphere.

The frustum is made by removing a small cone from a large cone.

The cones are similar.

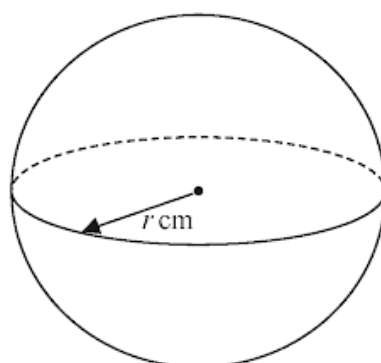
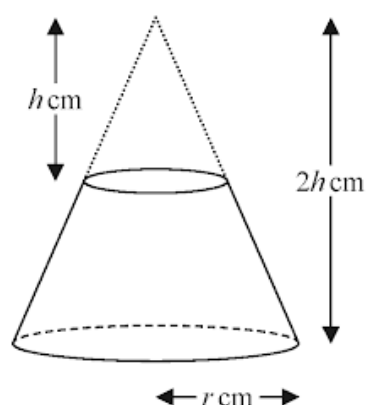


Diagram **NOT**
accurately drawn

The height of the small cone is h cm.

The height of the large cone is $2h$ cm.

The radius of the base of the large cone is r cm.

The radius of the sphere is r cm.

Given that the volume of the frustum is equal to the volume of the sphere,

find an expression for r in terms of h .

Give your expression in its simplest form.

The diagram shows two right-angled triangles, DEF and EFG .

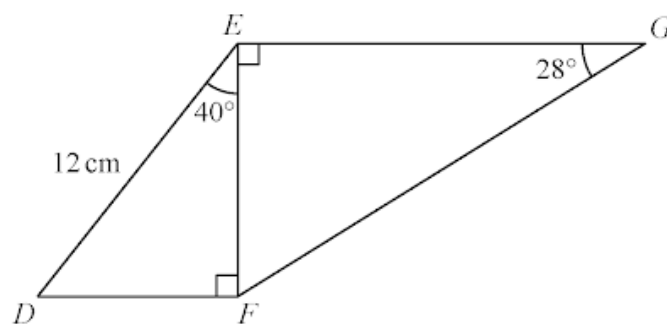


Diagram **NOT**
accurately drawn

Work out the length of EG .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question is 4 marks)

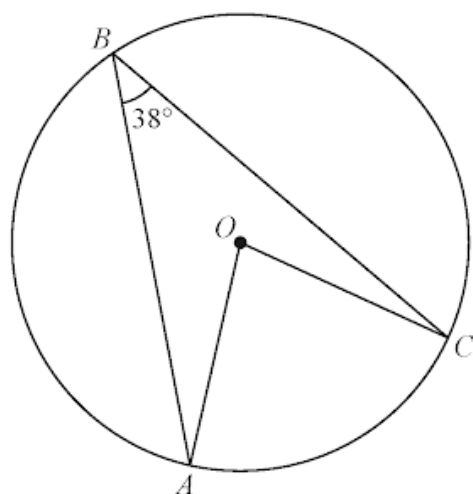


Diagram **NOT**
accurately drawn

A , B and C are points on a circle, centre O .

Angle $ABC = 38^\circ$

Work out the size of angle OAC .

Give a reason for each stage of your working.

(Total for Question is 4 marks)

The diagram shows a regular hexagon, $ABCDEF$, and an isosceles triangle, GHI .

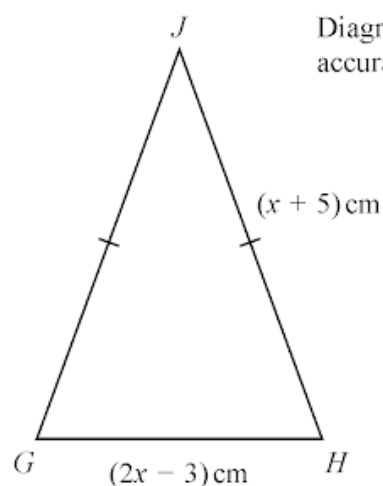
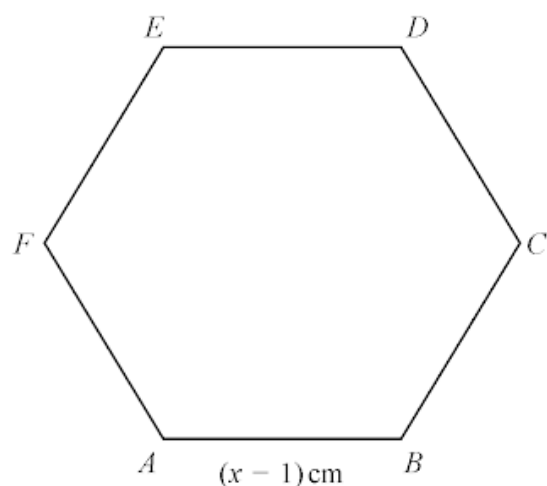


Diagram **NOT**
accurately drawn

The perimeter of the hexagon is equal to the perimeter of the triangle.

Find the length of each side of the hexagon.
Show clear algebraic working.

..... cm

(Total for Question is 5 marks)

Pablo made a solid gold statue.

He melted down some gold blocks and used the gold to make the statue.
Each block of gold was a cuboid, as shown below.

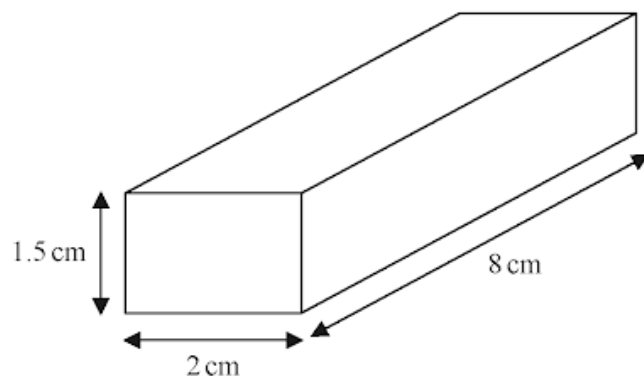


Diagram **NOT**
accurately drawn

The mass of the statue is 5.73 kg.
The density of gold is 19.32 g/cm^3

Work out the least number of gold blocks Pablo melted down in order to make the statue.
Show your working clearly.

(Total for Question is 5 marks)

The diagram shows a cuboid $ABCDEFGH$.

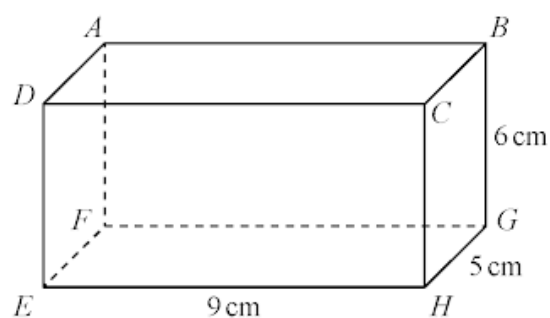


Diagram **NOT**
accurately drawn

$EH = 9\text{ cm}$, $HG = 5\text{ cm}$ and $GB = 6\text{ cm}$.

Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

o

(Total for Question is 4 marks)

The diagram shows parallelogram $EFGH$.

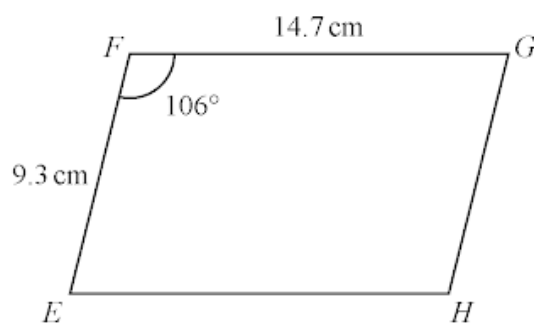


Diagram **NOT**
accurately drawn

$EF = 9.3$ cm
 $FG = 14.7$ cm
 Angle $EFG = 106^\circ$

- (a) Work out the area of the parallelogram.
 Give your answer correct to 3 significant figures.

..... cm²
 (2)

- (b) Work out the length of the diagonal EG of the parallelogram.
 Give your answer correct to 3 significant figures.

..... cm
 (3)

Here is a right-angled triangle.

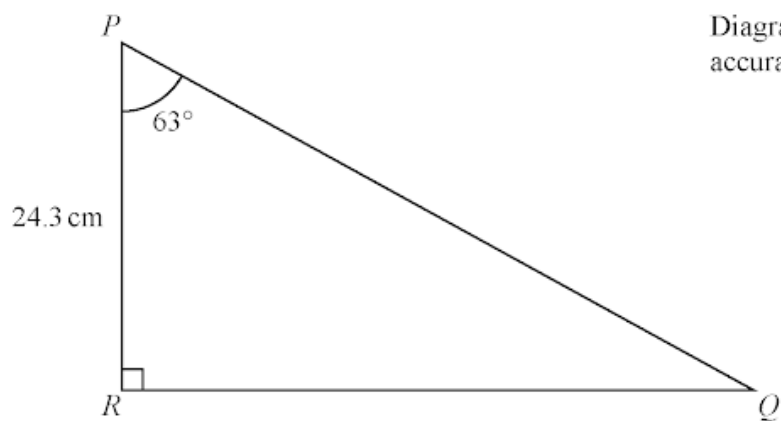


Diagram **NOT**
accurately drawn

Calculate the length of PQ .

Give your answer correct to 3 significant figures.

..... cm

The diagram shows a shape.

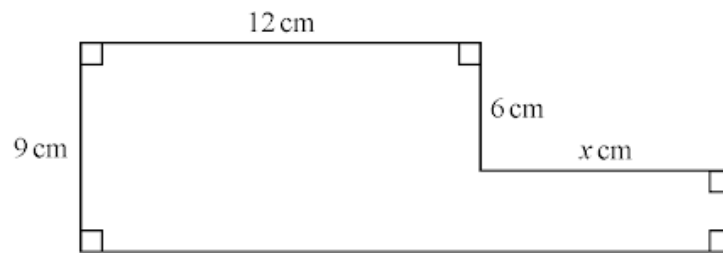


Diagram **NOT**
accurately drawn

The shape has area 129 cm^2

Work out the value of x .

$x =$

The diagram shows a regular octagon $ABCDEFGH$.

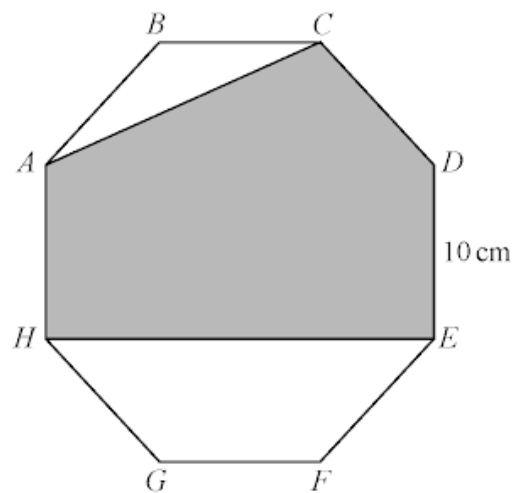


Diagram **NOT**
accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region $ACDEH$.
Give your answer correct to the nearest cm^2

The diagram shows a prism $ABCDEFGH$ with a horizontal base.

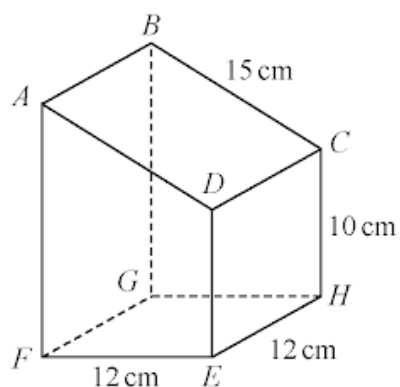


Diagram **NOT**
accurately drawn

The base of the prism, $EFGH$, is a square of side 12 cm.

Trapezium $ADEF$ is a cross section of the prism where AF and DE are vertical edges.

$$DE = CH = 10 \text{ cm}$$

$$AD = BC = 15 \text{ cm}$$

- (a) Work out the size of the angle between CF and the base $EFGH$.
Give your answer correct to one decimal place.

°

(3)

- (b) Work out the length of BE .
Give your answer correct to one decimal place.

cm

(3)

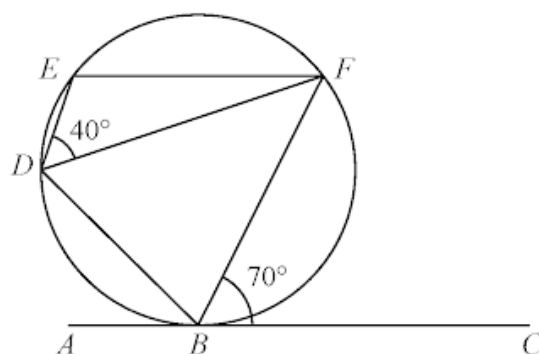


Diagram **NOT**
accurately drawn

B , D , E and F are points on a circle.
 ABC is the tangent to the circle at B .

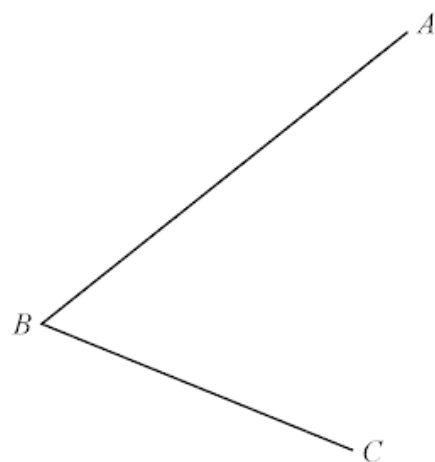
Angle $EDF = 40^\circ$

Angle $FBC = 70^\circ$

Prove that the tangent ABC is parallel to EF .
Give a reason for each stage of your working.

(Total for Question is 4 marks)

Using ruler and compasses only, construct the bisector of angle ABC .
You must show all your construction lines.



(Total for Question is 2 marks)

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R and **S** are two similar solid shapes.

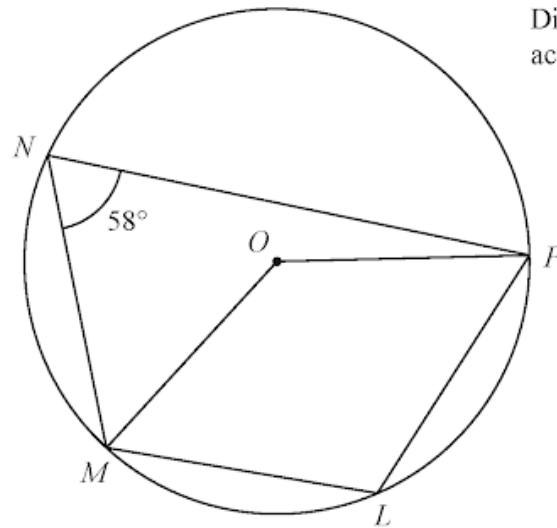
Shape **R** has surface area 108 cm^2 and volume 135 cm^3

Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

..... cm^3

(Total for Question is 3 marks)

Diagram **NOT**
accurately drawn

L, M, N and P are points on a circle, centre O

Angle $MNP = 58^\circ$

(a) (i) Find the size of angle MLP

○

(ii) Give a reason for your answer.

(2)

(b) Find the size of the reflex angle MOP

○

(2)

The diagram shows trapezium $ABCD$ in which BC and AD are parallel.

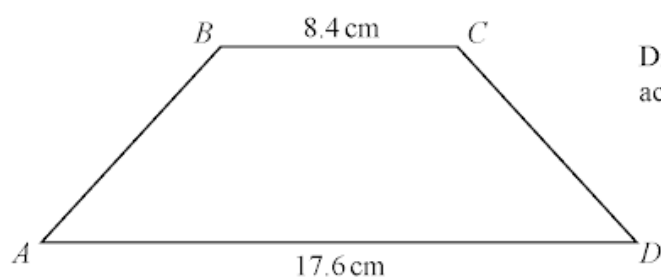


Diagram **NOT**
accurately drawn

The trapezium has exactly one line of symmetry.

$$BC = 8.4 \text{ cm}$$

$$AD = 17.6 \text{ cm}$$

The trapezium has area 179.4 cm^2

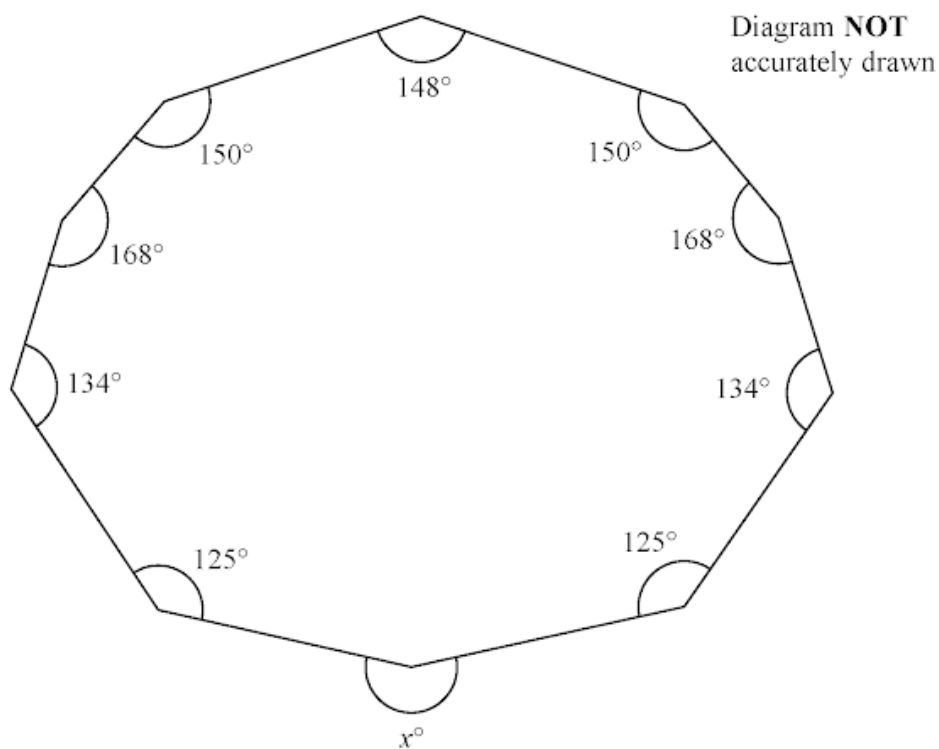
Work out the size of angle ABC .

Give your answer correct to 1 decimal place.

o

(Total for Question is 6 marks)

Here is a 10-sided polygon.



Work out the value of x .

$$x = \dots\dots\dots$$

(Total for Question is 4 marks)

Here is a triangle XYZ .

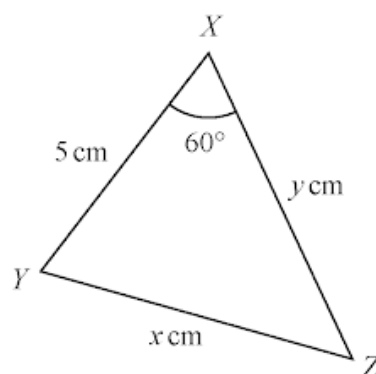


Diagram **NOT**
accurately drawn

The perimeter of the triangle is $k\text{ cm}$.

Given that $x = y - 1$

find the value of k .

Show your working clearly.

$$k =$$

(Total for Question is 5 marks)
